Sexual Reproduction in Flowering Plants

Comprehensive Study Notes

Stamen

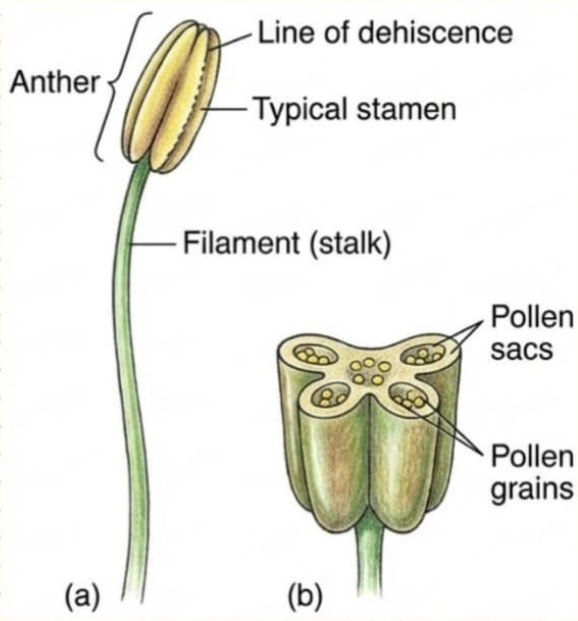
Stamen – the long and slender generally bilobed structure.

↳ {Attached to thalamus / petal}

1. Filament (stalk)
2. Anther
   - Bilobed
   - Dithecous lobe (long groove runs lengthwise to separate theca)

Structure:

- 4-sided tetra str. having 4 microsporangia located at corners, 2 in each lobe.
- It later turns to pollen sac.

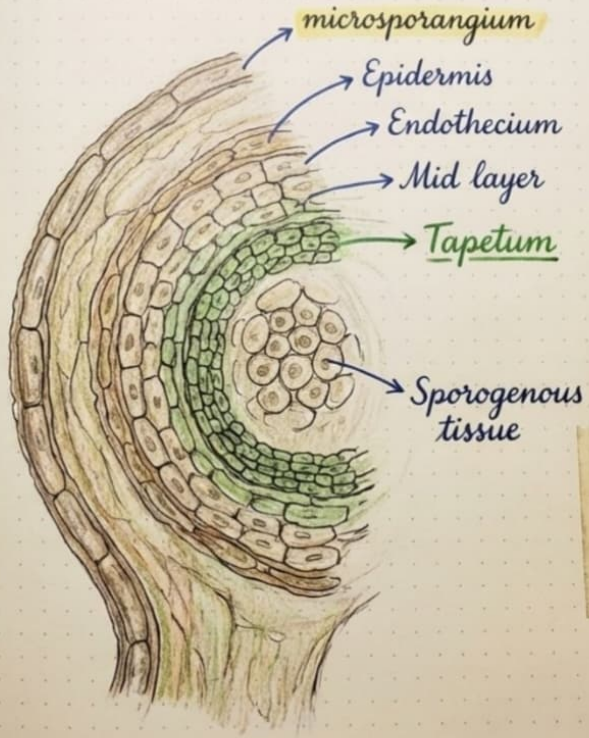


(a) A typical stamen;
(b) three-dimensional cut section of an anther

Structure of microsporangium:

- Has 4 layers:
  1. Epidermis
  2. Endothecium
  3. Mid layer
  4. Tapetum → dense cyto, multinucleated, provide nourishment.

- Sporogenous tissues are cells inside microsporangium.
- These spore cells undergo meiosis to form microspore tetrad.



Epidermis

Mid layer

microsporangium

Sporogenous tissues $\xrightarrow{\text{(meiosis)}}$ Microspore tetrad $\xrightarrow{\text{(microspore maturation)}}$ Pollen grains

The process of formation of microspores from a pollen mother cell (PMC) through meiosis is called microsporogenesis. As the anthers mature and dehydrate, the microspores dissociate from each other and develop into pollen grains.

Pollen grains : - 25 - 50 μm diameter.

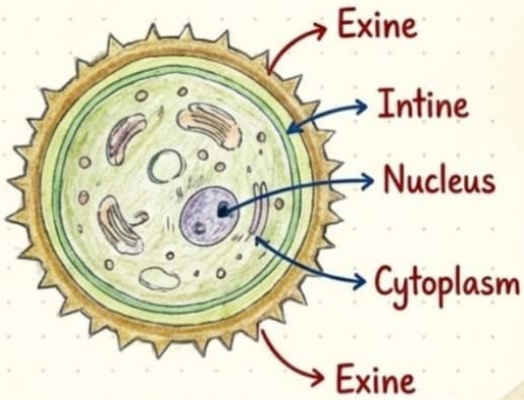
2 walls :

(i) Exine : outermost layer made of sporopollenin (most resist.).

- It can withstand high temp & strong acids + alkalis, thus pollen grains can be preserved as fossils.
- These pores where sporo is absent and facilitates pollen tube emerges.

(ii) Intine : thin, made of cellulose & pectin.

(iii) Cytoplasm : surr by plasma memb.



Pollen grain cells:

- Veg cell: biggest, food reserve, large irregular nucleus.
- Gen cell: small, floats in cyto, spindle-shaped, dense cytoplasm & nucleus.

→ 60%: shed at 2-celled stage (veg & gen)

→ 40%: 2 male gametes are formed (3-celled stage).

Pollen Viability:

- The period for which pollen grains remain viable is highly variable and to some extent depends on the prevailing temperature and humidity.
- In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release.
- In some members of Rosaceae, Leguminosae and Solanaceae, they maintain viability for months.

Pollen Banks:

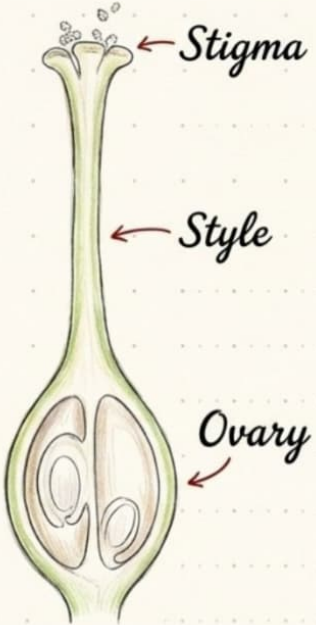
- It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-196°C). ❄️
- Such stored pollen can be used as pollen banks, similar to seed banks, in crop breeding programmes.

Pistil (Arrangement)

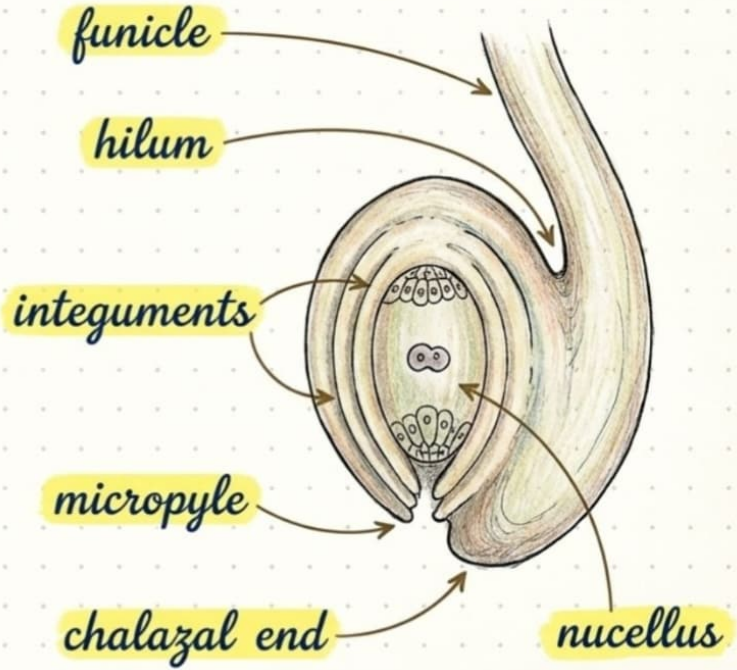
- → Monocarp (single pistil)
- → Multi pistil (multicarpellary)
  - Fusion: - Syncarpous (fused)
    - Apocarpous (free)

Pistil Parts

1. Stigma (landing platform for pollen grains)
2. Style (elongated slender part beneath stigma)
3. Ovary (basal bulged part)
   - → ovarian cavity / locule
   - → placenta present
   - → megasporangia / ovule arise → 1 or more than 1



- Ovule is attached to placenta by funicle.
- Ovule fuses with funicle through hilum.
- Each ovule has protective covering called integuments.
- Integuments leave a small opening called micropyle.
- Opp it has chalazal end.
- Inside integ. there is nucellus, having food reserves.



nucleus of this single megaspore

↓ (mitosis)

2-nucleus

↓

4-nucleus

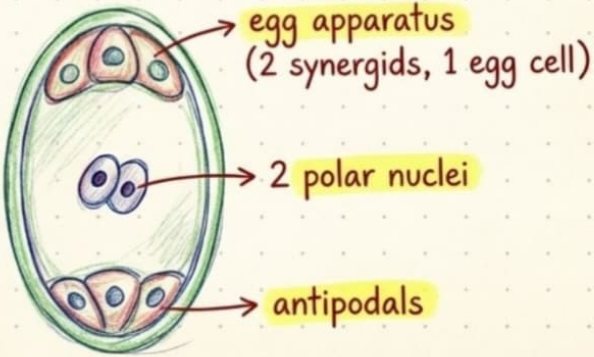
↓

8 nucleus

↓

Rearrangement:

- 3 cells are grouped at micropylar end constituting egg apparatus (2 synergids, 1 egg cell).
- 2 polar nuclei remain free in central cell.
- 3 cells move towards chalazal end & are called antipodals.



MMC (meiosis)

↓

4 megaspores formed

↓

only 1 remains functional

↓

embryo sac

↓

(monosporic development)

Pollination

Autogamy

- Pollination occurs in same flower.
- e.g.: Viola, oxalis, commelina.
  - → Chasmogamous (exposed stamen/pistil)
  - → Cleistogamous (closed flower)

Geitonogamy

- Pollination occurs among diff flowers but of same plant.

Xenogamy

- Pollination occurs among flowers of diff plants.

* Wind:

- Pollen grains must be light & non-sticky, and have well-exposed stamens & large feathery stigma (corn cob tassels).
- Have single ovuled ovaries - common in grass.

* Water:

- e.g.: Vallisneria, hydrilla (fresh water), Zostera (marine water).
- Pollen grains have mucilaginous covering for protection.
- Long, ribbon-like pollen grains.

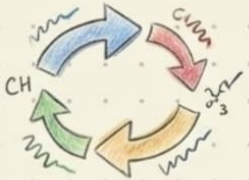
* Insects:

- Large, colourful, fragrant, rich in nectar flowers.
- Small flowers are arranged in clusters.
- Flies / beetles pollinated flowers have foul odours.
- Pollen grains stick to animal body.

Outbreeding Devices:

- Pollen release & stigma receptivity are non-sync.
- Anther & stigma are placed at diff positions.
- Self-incompatibility: It is a genetic mechanism & prevents self-pollen from fertilising ovule by inhibiting pollen germination / pollen tube growth in pistil.
- Prod. of unisex flowers.
  -> In castor & maize, unisex flowers are at same plant but in papaya flowers are at diff plants.

Pollen-pistil Interaction:



- Continuous dialogue b/w pollen & stigma by chem components.
- Pistil check if pollen is right type / wrong type.
- Promotes post-pollination events (pollen tube growth & seed germination).

Artificial hyb:

- One of major approach for crop improv programme.
- Emasculation is removal of anthers from flow bud before dehisces using forceps.
- These flowers are needed to be covered by butter paper called bagging.
- Once bagged flow attain receptivity, male gametes/pollen are dusted over stigma & rebagged & allowed for fruit development.

- PEC cells divides repeatedly to form primary endo tissue & are filled with food reserves & are used for nutrition of embryo.
- PEN undergoes division to give rise to free nuclear endosperm.
- The coconut water from tender coconut that you are familiar with, is nothing but free-nuclear endosperm and the surrounding white kernel is cellular endosperm.

Endosperm consumption:
- Completely consumed (e.g., pea, groundnut, beans) → non-albuminous / ex-albuminous
- Persists (e.g., castor, coconut, wheat, maize, barley) and be used up during seed germination → albuminous

Pollen tube Has 2 male gametes.

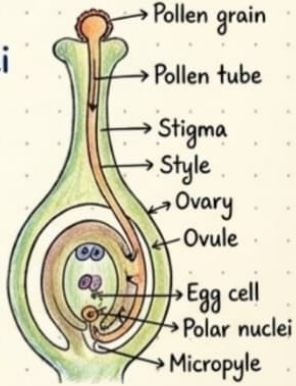
2 male gametes reach through pollen tube:

→ 1 goes towards egg cell & fuses with nucleus thus Syngamy occurs.

1 cell forms i.e zygote (2n).

→ other goes towards central cell & fuses with 2 polar nuclei thus triple fusion occurs.

3n formed i.e Primary endosperm cell (PEC) / Primary endosperm nucleus (PEN).



Dicot embryo	Monocot embryo
1. Embryonal axis & 2 cotyledons	1. Have 1 cotyledon
2. Above axis : epicotyl which terminates at plumule/stem tip Lower axis : hypocotyl terminates at radicle or root tip	2. It is called as scutellum i.e at one side of axis
	3. radicle at root tip called coleorhiza plumule at shoot tip called coleoptile

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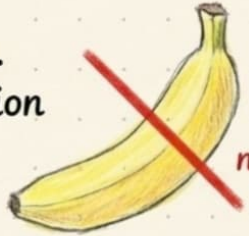
- The wall of the ovary develops into the wall of fruit called pericarp.

- True fruits: develop only from the ovary.
- False fruits: thalamus also contributes to fruit formation (apple, strawberry, cashew).
- Parthenocarpic fruits: fruits develop without fertilisation. Banana is one such example. Induced through the application of growth hormones and such fruits are seedless.



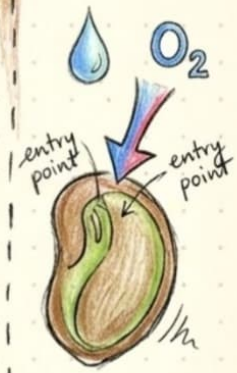
no seeds

Seeds advantages:

- Reproductive processes such as pollination and fertilisation are independent of water, seed formation is more dependable.
- Better adaptive strategies for dispersal to new habitats.

NotebookLM

- Integuments facilitates entry of water & O_2 into seed during germination.
- As seed matures, its moisture or water content reduces (10–15 per cent moisture by mass) & the metabolic activities slow down.
- & thus embryo enters into an inactive state called dormancy. If only favourable conditions are available then they will germinate.
- Perisperm: residual, persistent nucellus.



Apomixis:

- Produce seeds without fertilisation.
- Form of asexual reproduction that mimics sexual reproduction.
- Several ways of development of apomictic seeds. In some species, the diploid egg cell is formed without reduction division and develops into the embryo without fertilisation.

Polyembryony:

- Nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into the embryos (Citrus and Mango).
- Occurrence of more than one embryo in a seed is referred to as polyembryony.